Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

a.

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? ___ Serial0/0/0_ ___

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? ___129___

Does R2 show up as an OSPF neighbor on R1? ___Yes___

Does R2 show up as an OSPF neighbor on R3? ___No___

What does this information tell you?

R2 forms an OSPF neighbor relationship with R1 but not with R3 because the S0/0/1 interface on R2 is configured as a passive interface. Therefore, R3 reaches the 192.168.2.0/24 network through R1 instead of directly through R2._

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

router ospf 1

no passive-interface s0/0/1

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? ___ Serial0/0/1__________

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

The accumulated cost is 65. It is calculated by adding the cost of the serial link 64 from R3 to R2 and the GigabitEthernet interface cost 1 on R2.

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Is R2 listed as an OSPF neighbor to R3? __Yes______

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

The default OSPF reference bandwidth is 100 Mbps, which causes all links faster than 100 Mbps to have the same cost of 1. Changing the reference bandwidth allows OSPF to more accurately reflect the differences in link speeds such as Gigabit and 10 Gigabit Ethernet._

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Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

OSPF adds the cost of each interface along the path. The total cost to the network is the sum of all interface costs from the source router to the destination.

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i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

The new cost is 1562 because the bandwidth of the serial interfaces was changed to 128 Kb/s. This increased the OSPF cost of the interfaces, so the total path cost became higher.____

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

The route goes through R2 because the cost of the link to R3 was increased using the ip ospf cost command. OSPF chooses the path with the lowest cost, so R1 now uses the route through R2._

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Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   The router ID uniquely identifies a router in the OSPF network. It helps routers recognize each other and build OSPF neighbor relationships.

   2

2. Why is the DR/BDR election process not a concern in this lab?

   Because the routers are connected using point-to-point serial links. DR and BDR elections only happen on multiaccess networks like Ethernet._

   4

3. Why would you want to set an OSPF interface to passive?

   A passive interface stops sending OSPF hello messages. This reduces unnecessary routing traffic and improves security._

   4

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   OSPF uses areas. Dividing the network into areas makes the network easier to manage and more scalable.

   4

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   network 10.0.0.0 0.255.255.255 area 0

   4

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   The EIGRP route will be used because it has the lowest administrative distance compared to RIP and OSPF._

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